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Thermal hydraulic assessment of Once-Through Steam Generator

COURSE OF 'MULTIPHASE SYSTEMS AND TECHNOLOGIES' - MSc. OF NUCLEAR ENGINEERING

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Abstract: This is the project for the 2022-2023 course of '*Multiphase Systems and Technologies*'. This study focuses on the thermal-hydraulic analysis of a once-through steam generator (OTSG) in sodium-cooled fast reactors (SFRs). A reliable model was developed through Python code to evaluate heat transfer coefficients, heat transfer regimes, and pressure drops. The OTSG, critical for transferring heat from the sodium circuit to generate superheated steam, involves complex two-phase flow and heat transfer phenomena. The model was validated against higher fidelity code data, providing insights into heat transfer behavior and pressure drop characteristics. The script used for the analysis can be found at the following link: [OTSG code](#).

Key-words: Once Through Steam Generator, heat transfer coefficient, pressure drop, two-phase water.

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Nomenclature

Quantities		Nu	Nusselt number (–)
α	Heat transfer coefficient ($\frac{W}{mK}$)	p	Pressure (Pa)
Γ	Mass flow rate ($\frac{kg}{s}$)	Pe	Peclet number (–)
μ	Dynamic viscosity ($Pa\ s$)	Pr	Prandtl number (–)
ρ	Density ($\frac{kg}{m^3}$)	Q	Thermal power (W)
σ	Surface tension ($\frac{N}{m}$)	q	Heat surface rate ($\frac{W}{m^2}$)
Bo	Boiling number (–)	Re	Reynolds number (–)
C	Specific heat ($\frac{J}{kgK}$)	T	Temperature (K)
Co	Convective number (–)	v	Specific volume ($\frac{m^3}{kg}$)
D_h	Hydraulic diameter (m)	x	Equilibrium quality (–)
D_{int}	Internal diameter (m)	Subscripts	
D_{out}	External diameter (m)	b	Bulk
G	Mass flux ($\frac{kg}{m^2s}$)	g	Saturated steam
g	Gravity acceleration ($\frac{m}{s^2}$)	in	Inlet
h	Specific enthalpy ($\frac{J}{kg}$)	l	Saturated liquid
k	Thermal conductivity ($\frac{W}{mK}$)	Na	Sodium
L	Length (m)	out	Outlet
M	Mass (kg)	s	Wall surface
		w	Water

1. Introduction

The steam generator (SG) is a crucial element in Sodium-cooled Fast Reactors (SFRs). Its primary function is to transfer heat from the sodium loop to the steam/water loop, thus producing steam. This steam is then used to drive turbines for electricity generation. In Once-Through Steam Generators (OTSGs), feed water follows a continuous path, without the presence of steam drums or distinct sections for economizers, evaporators, and superheaters, resulting in the production of superheated steam at the outlet. SGs experience complex two-phase flow and heat transfer phenomena. The local thermal-hydraulic conditions within the tube bundle are critical, as they can lead to tube damage. Additionally, evaluating the local velocity and density conditions is essential for assessing the tube bundle design concerning flow-induced vibrations, which can cause fretting, wear, and fatigue of the tubes over the SG's design life. Therefore, thermal-hydraulic modeling of the SG is vital for its optimal design and safe operation.

The OTSG design examined in this study features an integrated vertical counter-current shell and tube heat exchanger (without baffles), with sodium on the shell side and feed water on the tube side, as depicted in Fig.1.

The tubes are straight, connected to the top and bottom tube sheets, and arranged in a triangular pitch with a tube support structure.

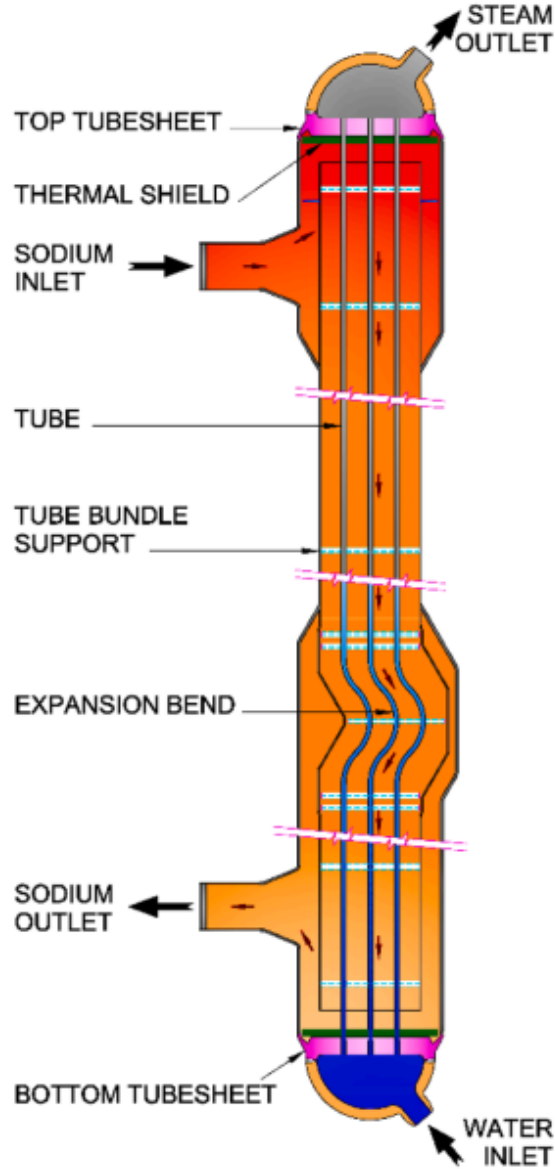


Figure 1: Sodium OTSG scheme

2. Objective

The goal of this project was to produce a realistic and faithful representation of the OTSG from the thermal-hydraulic point of view, able to replicate the results obtained by other scientific studies following higher degree models and using optimal thermal hydraulic codes [1].

3. Model

The conceptual modeling of the examined OTSG is here provided. First of all, no distinction has been made among the different tubes of the OTSG: all of them are assumed to be identical and to be characterized by the same thermal-hydraulic conditions. Hence, the analysis is carried out just on one representative tube and its surrounding channel.

The tube has been divided axially into 200 control volumes of equal height ΔL . Each of them comprehends 2 sides: the primary descending coolant, consisting of liquid sodium, and the secondary rising one, consisting of water. Each one of these control volumes is treated as an axial node of the system and the fluid properties and

balance equations were evaluated at each node. A schematic representation of the axial nodalisation is visible in Fig.2.

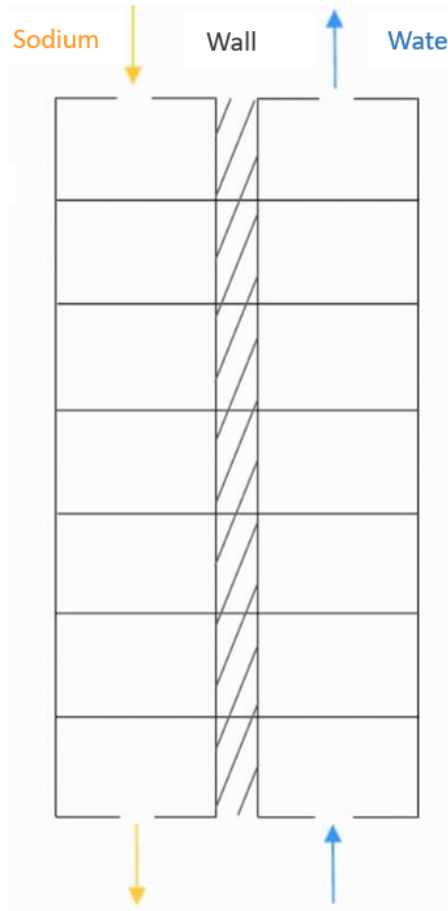


Figure 2: System model based on control volumes

The data regarding the OTSG geometry are presented in Tab.1

Geometry data	
Tube length	22 m
Tube inner diameter	12.6 mm
Tube outer diameter	17.2 mm
Shell inner diameter	831 mm
Number of tubes	547
Tube pitch (triangular)	32.2 mm
Tube roughness	30 μm
Tube material	9Cr-1Mo steel

Table 1: Geometrical data of the OTSG

The model made up for the resolution of this project is characterized by some simplifications of the actual OTSG design to guarantee a less computationally demanding and simpler simulation.

- Coolant properties and state variables were taken as homogeneous within every fluid volume. An arithmetic mean is carried out between the inlet and outlet of each CV to compute its average bulk temperature or specific enthalpy.
- The current analysis is mono-dimensional since only the axial coordinate is taken into account, whereas any radial discretization is discarded.

- The heat transfer across the shell towards the external environment is neglected, therefore the heat lost from the hotter sodium is entirely absorbed by the cooler water.
- Pressure drop of sodium is neglected since the variation between inlet and outlet is not significant and its properties are not relevantly affected by them.
- Radiative heat transfer from the tube's inner surface to the liquid core of the water flow when they are divided by a vapour blanket, following dryout, is neglected.
- Mass flow rates of both fluids are taken as constant along their flow, so there's no accumulation of mass within single CVs:

$$\frac{dM}{dt} = \Gamma_{out} - \Gamma_{in} = 0 \quad (1)$$

Where M is the fluid mass present in the considered CV.

- Localized pressure drops are neglected.

For each control volume, a system of equations was written, consisting of two energy equations (one per fluid volume):

$$\begin{cases} Q = \Gamma_{Na} C_{Na} (T_{Na,in} - T_{Na,out}) \\ Q = \Gamma_w (h_{w,out} - h_{w,in}) \end{cases} \quad (2)$$

one heat transfer equation through the wall:

$$Q = \frac{T_{Na,b} - T_{w,b}}{R_{th}} \quad (3)$$

where R_{th} is the thermal resistance between the bulk of the two fluids;

$$R_{th} = \frac{1}{2\pi\Delta L} \left(\frac{2}{\alpha_{Na} D_{int}} + \frac{1}{k_{wall}} \ln \left(\frac{D_{out}}{D_{int}} \right) + \frac{2}{\alpha_{Na} D_{out}} \right) \quad (4)$$

two equations regarding the heat transfer between each fluid and wall surface:

$$\begin{cases} q_{Na} = \alpha_{Na} (T_{Na,b} - T_{Na,s}) \\ q_w = \alpha_w (T_{w,s} - T_{w,b}) \end{cases} \quad (5)$$

The heat transfer coefficients are computed by exploiting the correlations displayed in the next section, chosen with respect to the particular heat transfer regime.

Last, one equation accounting for the pressure drops along the pipe:

$$\frac{dp}{dz} = \frac{dp_{fr}}{dz} + \frac{dp_{gr}}{dz} + \frac{dp_{ac}}{dz} \quad (6)$$

Where the three terms account for frictional, gravitational and acceleration pressure losses respectively.

4. Theoretical background

4.1. Flow boiling

Flow boiling is the process where a fluid moves across a heated surface, causing it to boil and evaporate. In this scenario, the fluid is not stagnant but flows over the surface, in this case due to a pump, which enhances the heat transfer due to the continuous movement and improved turbulence. As the fluid flows and receives heat from the surface, it reaches its boiling point, resulting in the formation of bubbles and vapor. During this process, the fluid experiences various flow regimes and heat transfer modes, each of which must be individually analyzed to better design the system.

4.1.1 Single phase liquid forced convection

The water enters the steam generator as subcooled liquid and exchanges heat with the hot surface by convection. The correlation chosen to describe this mechanism is the **Dittus-Boelter** one (Eq.7), developed for turbulent flows.

$$Nu = 0.023 Re^{0.8} Pr^{0.4} \quad (7)$$

Sodium is in the subcooled liquid state as well and remains in this condition throughout the entire steam generator. Therefore the only correlation used to compute its Nusselt number and then its heat transfer coefficient is calculated thanks to **Lyon's** correlation:

$$Nu = 7 + 0.025Pe^{0.8} \quad (8)$$

where:

$$Pe = RePr = \frac{GD_h}{\mu} \frac{C\mu}{k} = \frac{GD_h C}{k} \quad (9)$$

Sodium heat capacity and thermal conductivity are calculated as:

$$\begin{aligned} C_{Na} &= 1000(1.43674 - 5.8049e-4 T + 4.6229e-7 T^2) \\ k &= 94 - 3.25e-2(1.8 T + 32) + 3.62e-6(1.8 T + 32)^2 \end{aligned}$$

4.1.2 Onset of nucleate boiling

The onset of nucleate boiling (ONB) marks the transition from single-phase liquid convection to a combination of convection and nucleate boiling, characterized by the formation of vapor bubbles. Nucleation is divided into two categories: homogeneous, occurring within a superheated liquid mass, and heterogeneous, occurring at a solid-liquid interface. In the current scenario, nucleation happens heterogeneously due to surface imperfections and contaminants, which trap vapor/gas in cavities. These nucleation sites enhance the formation of bubbles, lowering the required superheat for their formation compared to theoretical homogeneous nucleation. Heterogeneous nucleation is easier because vapor/gas trapped in surface cavities can grow if the surrounding fluid temperature exceeds the vapor's saturation temperature by an amount suggested by **Frost-Dzakowic** correlation:

$$\Delta T_{s,ONB} = \sqrt{\frac{8\sigma T_{sat} q}{k_l h_{lg} \rho_g}} Pr_l = T_{s,ONB} - T_{sat} \quad (10)$$

This formula takes the work done by **Davis-Anderson** for pool boiling and adapts it to flow boiling by multiplying for Pr_l , that is Prandtl's number associated to liquid properties. This way I can account for the thinner thermal boundary layer present in this case, due to the larger velocity and heat transfer capacity characteristic of flow boiling, and its perturbation caused by emerging bubbles.

4.1.3 Subcooled nucleate boiling

Once boiling is started and the temperature of bulk of the fluid is still under the saturation one, the heat transfer regime of subcooled nucleate boiling begins. As the vapor bubbles grow and detach from the heated surface, they move into the cooler bulk liquid and they quickly collapse and condense back into the liquid state. This rapid condensation contributes to intense localized heat transfer. To investigate this region, **Rohsenow's** approach is adopted:

$$\begin{aligned} q_{lo} &= \alpha_{lo}(T_s - T_b) \\ q_{ONB} &= C_{NB}(T_{s,ONB} - T_{sat}) \\ q_{NB} &= C_{NB}(T_s - T_{sat}) \\ q^2 &= q_{lo}^2 + (q_{NB} - q_{ONB})^2 \end{aligned}$$

Where C_{NB} is defined as:

$$C_{NB} = \mu_l h_{lg} \left(\frac{C}{C_{sf} h_{lg} Pr_l} \right)^3 \sqrt{\frac{g(\rho_l - \rho_g)}{\sigma}} \quad (11)$$

While C_{sf} , experimental constant depending on fluid-material combination, is taken as 0.013, given the lack of information about the interaction between these two components. The aforementioned correlation describes heat transfer in this region as a superposition of convective and boiling phenomena. As the wall temperature raises and more nucleation sites are activated, bubbles formation increases and the latter mechanism becomes more and more relevant.

Eventually the heat transfer coefficient can be simply evaluated from the heat transfer equation:

$$\alpha_{SCB} = \frac{q}{T_s - T_b} \quad (12)$$

4.1.4 Saturated nucleate boiling

As soon as the bulk temperature of the fluid becomes equal to the saturation one, the saturated nucleate boiling regime begins. In the initial phase, heat transfer is primarily influenced by the extent of bubble formation on the solid surface. As long as nucleation occurs, it significantly impacts the heat transfer rate. When the flow quality is high, annular flow develops, causing the liquid film to thin due to evaporation and droplet entrainment. The heat transfer from the liquid film to the vapor core becomes so effective that nucleation within the film may be suppressed. Consequently, evaporation mainly takes place at the liquid film–vapor interface. These two contributions are taken into account within the two terms that constitute **Kandlikar's** correlation [2]:

$$\alpha_{SB} = (C_1 Co^{C_2} + C_3 Bo^{C_4}) \alpha_{lo} \quad (13)$$

Where α_{lo} is the heat transfer coefficient as if only the liquid was flowing across the tube with a mass flux equal to the one of the entire mixture, obtained from Dittus-Boelter. The values of the four constants are listed in Tab.2.

Constants	Convective region	Nucleate boiling region
C_1	1.136	0.6683
C_2	-0.9	-0.2
C_3	667.2	1058
C_4	0.7	0.7

Table 2: Constants of Kandlikar's correlation

When α_{NB} computed using the constants of the first column is greater than the one obtained with the ones of the second column, then the regime is considered to be convective dominant and the first value of the heat transfer coefficient is exploited. In the opposite case, the heat transfer regime is seen as boiling dominant and the second value is taken:

$$\alpha_{NB} = \begin{cases} \alpha_{CD}, & \alpha_{CD} \geq \alpha_{NBD} \\ \alpha_{NBD}, & \alpha_{CD} < \alpha_{NBD} \end{cases} \quad (14)$$

4.1.5 Critical Heat Flux and post dryout

Critical Heat Flux (CHF) represents the maximum crucial heat flux that leads to the transition from nucleate boiling to a less efficient regime, where walls are no more in contact with a liquid but just with vapor, although the fluid is not fully evaporated yet.

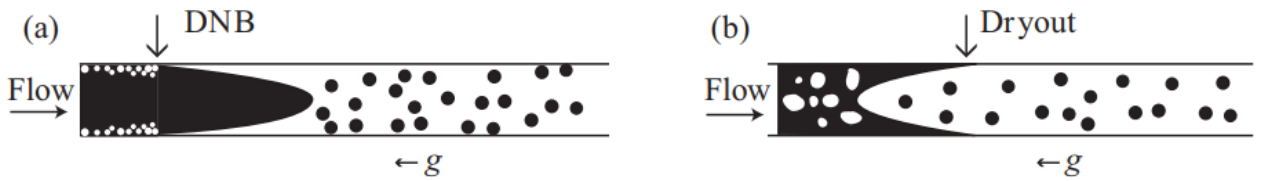


Figure 3: DNB and dryout schematics

Two types of thermal crisis can be distinguished in flow boiling: Departure from Nucleate Boiling (DNB), typical of low quality mixtures, and Dryout, characteristic of higher quality ones. These two phenomena are shown in Fig.3. The former occurs when the heat flux becomes so large, that the number of nucleation sites and so of bubbles close to the wall becomes so high that the liquid is prevented from reaching the surface. This brings to the formation of a vapour layer attached to the wall, entraining the remaining liquid along the pipe axis. Dryout instead, typical in channels where annular flow is dominant, occurs when the liquid film in contact with the heated wall completely evaporates, leading to a boiling crisis. The liquid film is replenished by the deposition of droplets from the vapor core but is simultaneously depleted by droplet entrainment caused by the high-velocity vapor shearing off liquid and forming surface waves. Additionally, some liquid evaporates directly at the film-vapor interface. Dryout, happens when the rate of liquid film depletion surpasses the replenishment rate, causing the film to evaporate entirely. This exposes the heated surface directly to the vapor, which

drastically reduces heat transfer efficiency and can lead to overheating.

The **2006 Groeneveld Look-Up Table (LUT)** [3] are exploited for predicting the value of CHF. This database compiles extensive experimental evidence to provide reliable CHF values for a wide range of pressures, mass fluxes, and qualities, evaluated for a hydraulic diameter of 8 mm. Through interpolation, an estimate is extracted from this database and then corrected to account for the diameter of the duct in the current scenario:

$$q_{CHF}(D_h, p, G, q) = q_{CHF}(D = 8 \text{ mm}, p, G, q) \left(\frac{D_h (\text{mm})}{8 \text{ mm}} \right)^{-0.5} \quad (15)$$

To characterize the heat transfer coefficient in the post-dryout regime, the **Dougall-Rohsenow's** correlation is exploited:

$$Nu = \frac{\alpha_{PD} L_c}{k_g} = a \left(Re_g \left(x + \frac{\rho_g}{\rho_l(1-x)} \right) \right)^b Pr_g^c \quad (16)$$

Where the constants shown are listed in Tab.3.

Constants	
a	0.023
b	0.8
c	0.4

Table 3: Constants of Dougall-Rohsenow's correlation

This correlation is based on Dittus-Boelter's one and considers the two phases in thermal equilibrium and the main heat mechanism is assumed to be between the wall and saturated steam.

4.1.6 Single phase vapor forced convection

Once the fluid has become saturated vapor, the heat transfer regime evolves into gaseous forced convection, described again by Dittus-Boelter 7.

4.2. Pressure drops

A fluid flowing across a pipe is subjected to pressure loss, its mechanical energy is converted into other forms, such as potential one and heat. The mathematical approach to predict this phenomenon relies on the balance of momentum. The contributions to this energy conversion are different in case the fluid is streaming as single- or two-phase flow Three different contributions can be distinguished:

- Frictional, accounting for the fluid-structure interaction between the two-phase flow and the bounding channel wall. This term is always negative for co-current two-phase flows and represents pressure energy degradation due to viscous dissipation within the flow.
- Gravitational, accounting for the movement of the two-phase flow within the gravitational force field. It is negative if the flow is upwards, indicating that pressure energy is converted into potential gravitational energy.
- Accelerating, accounting for the acceleration or deceleration of the two-phase flow along the channel due to density variations and the conversion of kinetic and pressure energies from one into the other.

4.2.1 Single phase flow

In this case, the three pressure drops along a single control volume are:

- Frictional:

$$\frac{dp_{fr}}{dz} = - \frac{f G^2}{2 \rho D_{int}} \quad (17)$$

Darcy's friction factor is computed by exploiting **Blasius'** correlation:

$$f = \frac{0.316}{Re^{0.25}} \quad (18)$$

- Gravitational:

$$\frac{dp_{gr}}{dz} = -g\rho \quad (19)$$

- Acceleration:

$$\frac{dp_{ac}}{dz} = -G^2 \frac{d\left(\frac{1}{\rho}\right)}{dz} \quad (20)$$

4.2.2 Two phase flow

In this scenario, the fluid is characterized by two different phases having different features and they both concur to the loss of momentum of the mixture. Two models have been developed to approach this scenario: homogeneous or separated flows. The former treats the mixture as a single fluid with average properties and assumes that the gaseous and liquid phases are flowing at the same speed. The latter instead looks at the mixture as two independent fluids flowing at different velocities and associates their single pressure losses to the one of the solution. In this analysis, the separated flow model has been chosen.

- Frictional: **Muller-Steinhagen and Heck's** correlation is used:

$$\frac{dp_{fr}}{dz} = \left(\frac{dp_{lo}}{dz} + 2\bar{x} \left(\frac{dp_{go}}{dz} - \frac{dp_{lo}}{dz} \right) \right) (1 - \bar{x})^{1/3} + \frac{dp_{go}}{dz} \bar{x}^3 \quad (21)$$

Where the liquid-only and gas-only pressure gradients are:

$$\frac{dp_{lo}}{dz} = \frac{f_l G^2}{2\rho_l D_{int}} \quad \frac{dp_{go}}{dz} = \frac{f_g G^2}{2\rho_g D_{int}} \quad (22)$$

The single-phase Darcy's friction factors are again estimated through **Blasius'** correlation:

$$f_l = \frac{0.316}{Re_{lo}^{0.25}} \quad f_g = \frac{0.316}{Re_{go}^{0.25}} \quad (23)$$

Where Reynold's number in case of liquid-only and gas-only conditions are defined as:

$$Re_{lo} = \frac{GD_{int}}{\mu_l} \quad Re_{go} = \frac{GD_{int}}{\mu_g} \quad (24)$$

- Gravitational:

$$\frac{dp_{gr}}{dz} = -g\bar{\rho}^* \quad (25)$$

Where $\bar{\rho}^*$ is the actual density:

$$\bar{\rho}^* = \rho_g \bar{\epsilon} + \rho_l (1 - \bar{\epsilon}) \quad (26)$$

The average void fraction of the pipe cross-section $\bar{\epsilon}$ is computed by exploiting **Bankoff's** correlation:

$$\bar{\epsilon} = K_b \bar{\epsilon}_h \quad (27)$$

Where K_b is a constant calculated as:

$$K_b = 0.71 + 0.0145 p \text{ (MPa)} \quad (28)$$

While $\bar{\epsilon}_h$ is the average void fraction in case of homogeneous flow:

$$\bar{\epsilon}_h = \left(1 + \frac{\rho_g}{\rho_l} \frac{1 - \bar{x}}{\bar{x}} \right)^{-1} \quad (29)$$

- Acceleration:

$$\frac{dp_{ac}}{dz} = -G^2 \frac{d\bar{v}_m}{dz} \quad (30)$$

Where $\bar{v}_m$ is the momentum averaged specific volume, defined as:

$$\bar{v}_m = v_g \frac{\bar{x}^2}{\epsilon} + v_l \frac{(1 - \bar{x})^2}{1 - \epsilon} \quad (31)$$

5. Resolution model

The code is elaborated in this way, as shown in Fig.4: starting from the lowest control volume, the two wall surface temperatures $T_{s,Na}$ and $T_{s,w}$ are initially supposed as a guess, starting from them, the total power exchanged is computed through Eq.32. Water properties are taken from IAPWS-97 [4], exploiting the Python package pyXSteam [5].

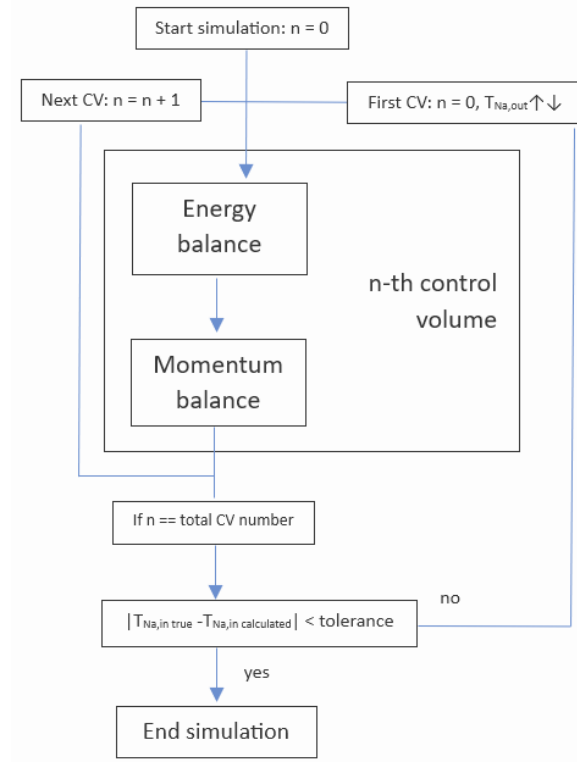


Figure 4: Code structure

The inlet water temperature and outlet sodium one are the two necessary boundary conditions of the system, although the latter is not known yet, so an initial guess is proposed and will be adjusted only afterward. A schematic control volume is shown in Fig.5

$$Q = \frac{2\pi k \Delta L (T_{s,Na} - T_{s,w})}{\log(D_{out}/D_{int})} \quad (32)$$

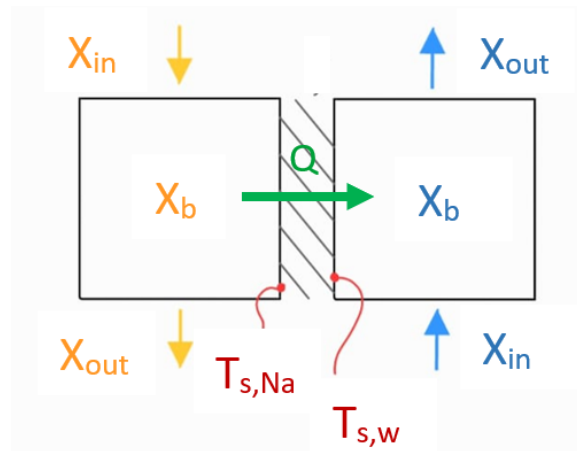


Figure 5: CV schematic. X is a generic quantity or property of the two fluids acquiring different values while flowing across the CV.

At this point, the temperature of sodium at the inlet and the specific enthalpy of water at the outlet of the

CV taken into account are computed, exploiting the energy balance on the two coolants, as done in Eq.33. All water properties are evaluated at the inlet pressure, since the average one is still unknown and its variation is not supposed to affect significantly the result.

$$\begin{cases} T_{Na,in} = T_{Na,out} + \frac{Q}{\Gamma_{Na} C_{Na}} \\ h_{w,out} = h_{w,in} + \frac{Q}{\Gamma_w} \end{cases} \quad (33)$$

Now the average bulk temperatures of the two fluids are calculated through an arithmetical mean between the inlet and outlet ones, as shown in Eq.34.

$$\begin{cases} T_{Na,b} = \frac{T_{Na,in} + T_{Na,out}}{2} \\ h_{w,b} = \frac{h_{w,in} + h_{w,out}}{2} \\ T_{w,b} = T@ (h_{w,b}, p_w) \end{cases} \quad (34)$$

Where water temperature is assessed from the knowledge of its specific enthalpy and pressure through IAPWS-97 tables [4].

The heat surface rates together with the heat transfer coefficients related to each wall side are then evaluated through Eq.35.

$$\begin{cases} q_{Na} = \frac{Q}{\pi D_{out} \Delta L} \\ q_w = \frac{Q}{\pi D_{int} \Delta L} \\ \alpha_{Na} = HTC(T_{Na,b}, G_{Na}, D_{h,Na}) \\ \alpha_w = HTC(p_w, h_{w,b}, T_{s,w}, q_w, G_w, D_{h,w}) \end{cases} \quad (35)$$

The ratio between the heat surface rates and the product between the heat transfer coefficients and the temperature difference between the fluids and the walls, as established by the heat transfer equation, should be unitary. Therefore the numerical solver is requested to determine the residual of these two equations, as shown in Eq.36, and check if it fulfills the demanded tolerance. If not, other guesses of the wall surface temperatures are taken and an iterative process is started.

$$\begin{cases} residual_{Na} = \frac{q_{Na}}{\alpha_{Na}(T_{Na,b} - T_{s,Na})} - 1 \\ residual_w = \frac{q_w}{\alpha_w(T_{s,w} - T_{w,b})} - 1 \end{cases} \quad (36)$$

Now, once the thermal aspect has been dealt with, the pressure drops are calculated. Depending on the state of the system, the single-phase governing equations or the two-phase ones are exploited and the discrete pressure losses related to frictional, gravitational, and acceleration contributions are evaluated. The first two are computed through trapezoidal integration between the inlet and the outlet of the CV. In the simpler case of single-phase flow the pressure losses across the CV are:

$$\Delta p_{fr} = - \int_{z_{in}}^{z_{out}} \frac{f G^2}{2 \rho D_{int}} dz = - \frac{G^2}{4 D_{int}} \left(\frac{f_{in}}{\rho_{in}} + \frac{f_{out}}{\rho_{out}} \right) \Delta L \quad (37)$$

$$\Delta p_{gr} = - \int_{z_{in}}^{z_{out}} g \rho dz = - g \frac{\rho_{in} + \rho_{out}}{2} \Delta L \quad (38)$$

$$\Delta p_{ac} = - \int_{z_{in}}^{z_{out}} \frac{d \left(G^2 \frac{1}{\rho} \right)}{dz} dz = - G^2 \int_{z_{in}}^{z_{out}} d \left(\frac{1}{\rho} \right) = - G^2 \left(\frac{1}{\rho_{out}} - \frac{1}{\rho_{in}} \right) \quad (39)$$

The pressure of water exiting the CV is then given by:

$$p_{out} = p_{in} + \Delta p = p_{in} + (\Delta p_{fr} + \Delta p_{gr} + \Delta p_{ac}) \quad (40)$$

Once the system has been solved at the initial lowest control volume, the outlet water temperature and the inlet sodium one are known and exploited as boundary conditions for the next upper volume. This procedure is carried on until the top volume is reached and dealt as well. At this point, the final inlet sodium temperature which has been computed is compared with the actual one. If the outcome is sufficiently close to the real one, the resolution stops, otherwise another guess for the outlet sodium temperature of the lowest volume is taken and the process starts again. In particular, if the computed temperature is hotter than the actual one, a colder guess will be chosen and vice versa in the opposite case.

6. Results

The outcome of the simulation is presented below. First, fluids' temperatures and heat transfer coefficients are shown in Fig.6 and Fig.7 respectively. Water enters the OTSG as subcooled and heats up along the pipe until around meter 4, where the wall surface becomes sufficiently superheated that boiling is initiated, nucleation sites are activated and the heat transfer coefficients start rising. This reduces the temperature gap between water and the wall surface, due to the contraction of the thermal resistance, and increases the exchanged power. Once the saturation condition is reached, this behavior is even more emphasized and the temperature of the surface of the wall next to the evaporating water remains almost constant.

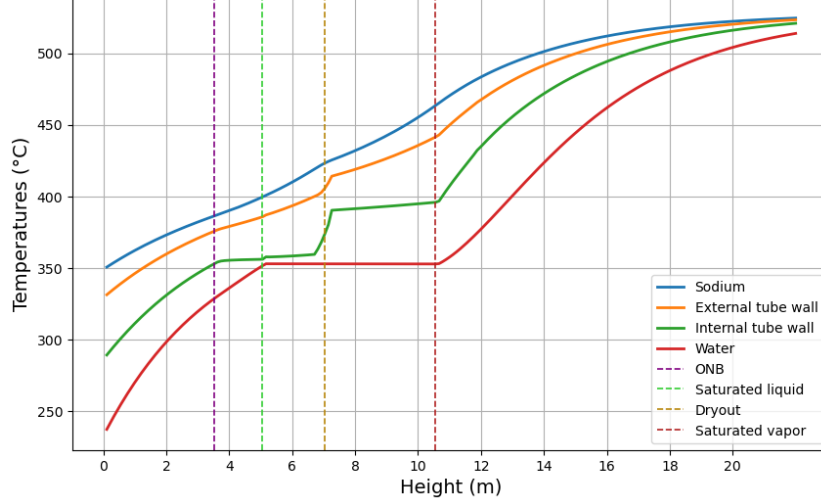


Figure 6: Temperature profiles of liquid sodium, water and both wall surfaces along the pipe length

At a certain point, around meter 7, the surface heat transfer reaches the CHF value and dryout occurs. The liquid film flowing in contact with the wall and wetting it is depleted, although complete evaporation of the fluid has not occurred yet, and the heating surface is now touched just by vapor. This causes the heat transfer coefficient and the exchanged power to drop abruptly, while the side of the wall in contact with water promptly warms up. The remaining liquid keeps evaporating in the pipe through the heat received by the steam till saturation condition is achieved. In the meantime, the heat transfer coefficient rises till saturation, due to the increasing velocity and turbulence of the fluid and the presence of liquid droplets inside it. Once all the liquid phase has evaporated, the fluid keeps warming up becoming superheated vapor. Its heat transfer coefficient decreases more and more and so does the temperature difference while it approaches the one of sodium at its inlet.

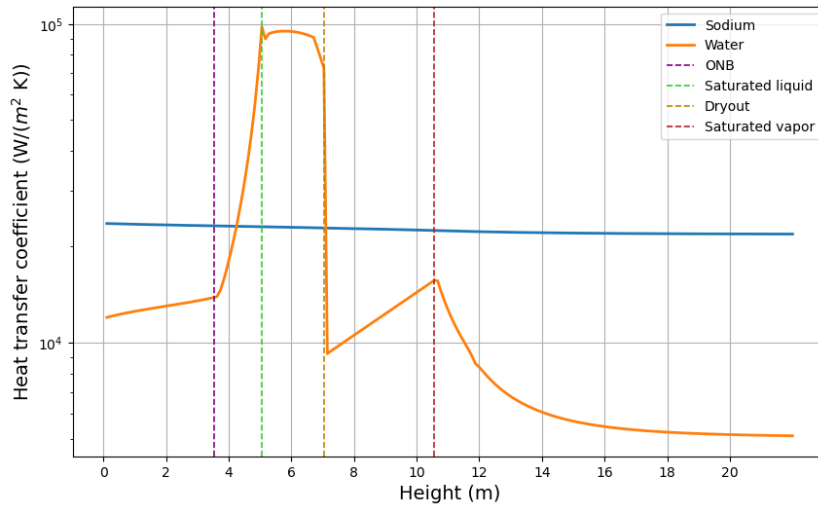


Figure 7: Heat transfer coefficients of both liquid sodium and water along the pipe length

Regarding pressure drops instead, as it can be seen in Fig.8, the gravitational term is the dominant one in

the bottom part of the system, where the density of the fluid is at its peak, and descends going upward. The frictional term, on the contrary, increases due to the fluid velocity rise, which opposes to the density reduction in order to fulfill the balance of mass. Last, the acceleration term grows till the vapor saturation condition, due to the fact that flow velocity and density variation within each CV are significant, while it decreases more and more once the fluid is only gaseous and these gradients become less relevant.

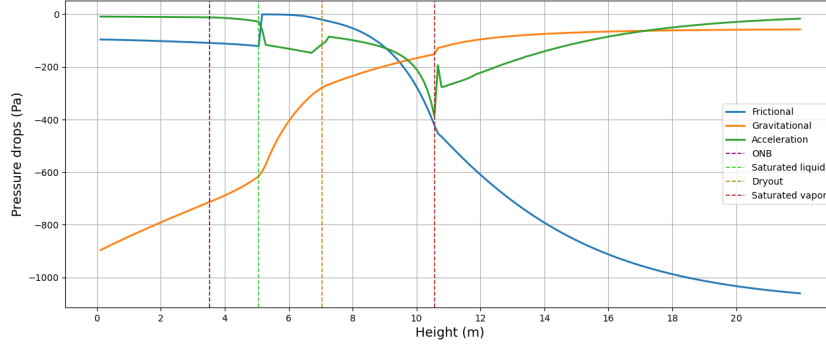


Figure 8: Pressure drops along the pipe length

7. Sensitivity Analysis

A sensitivity analysis regarding 5 significant parameters has been conducted to investigate their influence on the system behavior. These 5 quantities are:

- Water mass flow rate Γ_{water}
- Water inlet pressure $p_{in,water}$
- Water inlet temperature $T_{in,water}$
- Sodium mass flow rate Γ_{Na}
- Sodium inlet temperature $T_{in,Na}$

Each of them is explored at 4 different values with respect to the nominal ones: 90%, 95%, 105% and 110%.

7.1. Water mass flow rate

The results obtained for different values of the water mass flow rate are presented in Fig.9 and Fig.10.

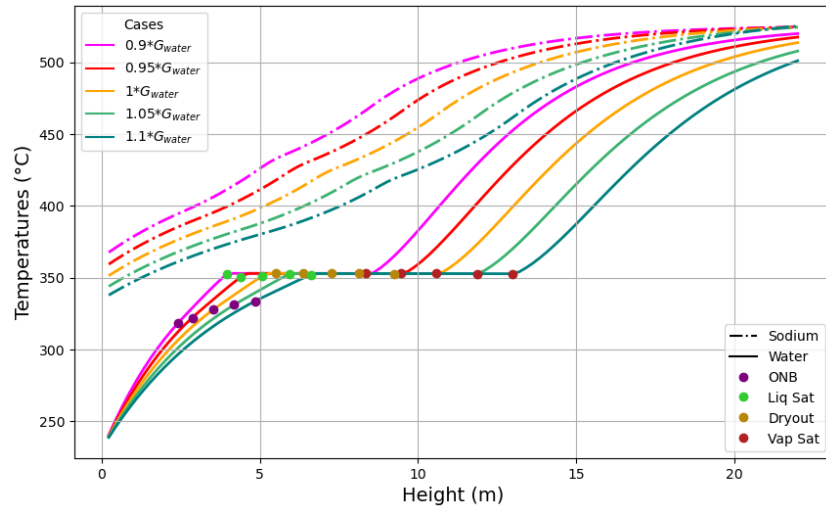


Figure 9: Temperature profiles of liquid sodium and water in case of different water mass flow rates

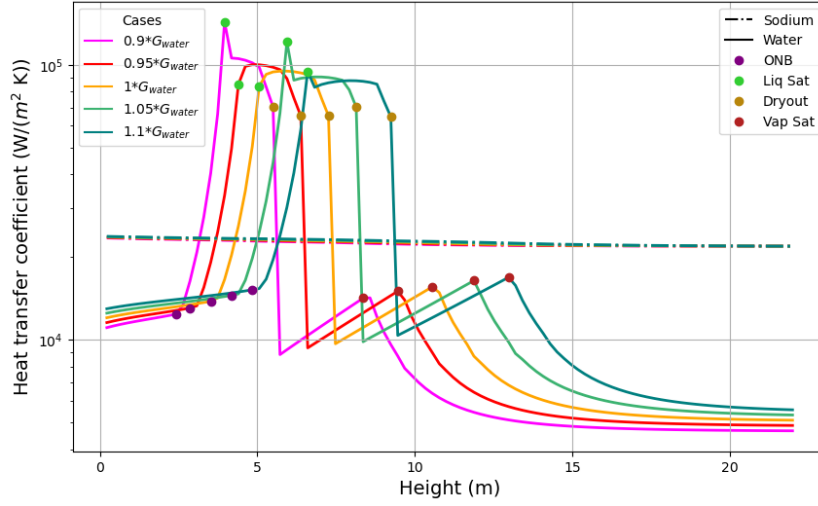


Figure 10: Heat transfer coefficients of liquid sodium and water in case of different water mass flow rates

Clearly lower mass flow rates shift the fluid temperature plots to the left, since a lower thermal capacity of water leads it to warm up sooner, while sodium is cooled less. For this same reason, water remains less time in the saturation region since it takes less heat to evaporate completely. The heat transfer coefficients is affected as well. At lower mass flow rates, turbulence is reduced and so does also the capability of water to exchange heat. Within the saturation region instead, due to the fact that sodium is hotter at lower water mass fluxes, the temperature difference between the two fluids is higher and so also the heat surface rate and the Boiling number.

7.2. Water inlet pressure

Reducing water inlet pressure leads saturation temperature to decrease, therefore onset of boiling and saturation conditions are anticipated, while the heat required for evaporation grows, as shown in Fig.11.

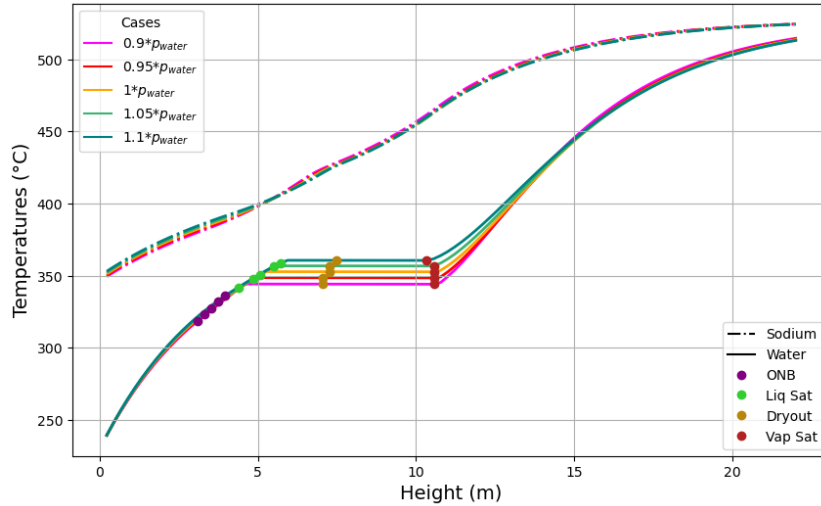


Figure 11: Temperature profiles of liquid sodium and water in case of different water inlet pressures

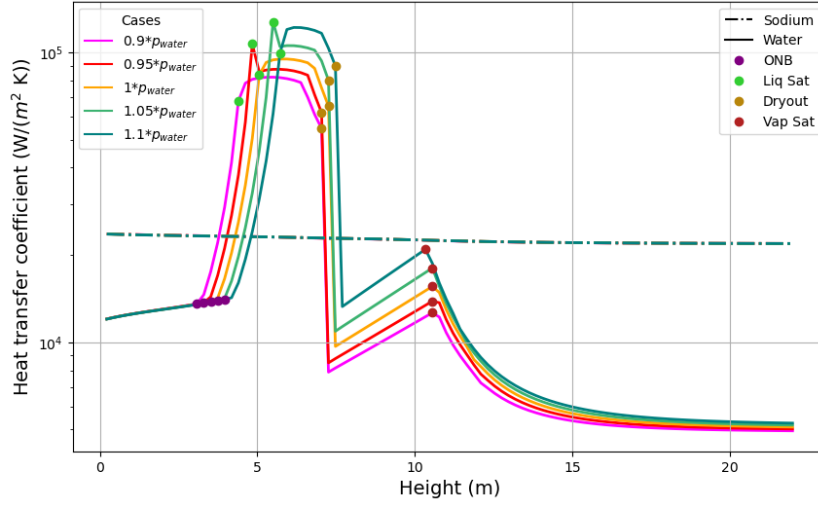


Figure 12: Heat transfer coefficients of liquid sodium and water in case of different water inlet pressures

Regarding the heat transfer coefficient instead, visible in Fig.12, lower pressure, as already said, brings to larger heat of evaporation, therefore boiling phenomenon requires more energy to be sustained. This hinderance carries to a higher heat transfer coefficient in case of larger pressures, where boiling, and its positive advantages to heat exchange, can be exploited more.

7.3. Water inlet temperature

As it can be seen in Fig.13, reducing the water inlet temperature leads the temperature plot to shift right, since now water is colder on average, therefore it is able to subtract more energy from sodium. Transitions are first met by hotter water flows. Since they are less able to cool the sodium, they encounter a warmer flow, and so also a warmer wall surface, which initiates nucleate boiling earlier. Saturation instead is met sooner because hotter water flows are less subcooled, so they require less energy to reach saturation.

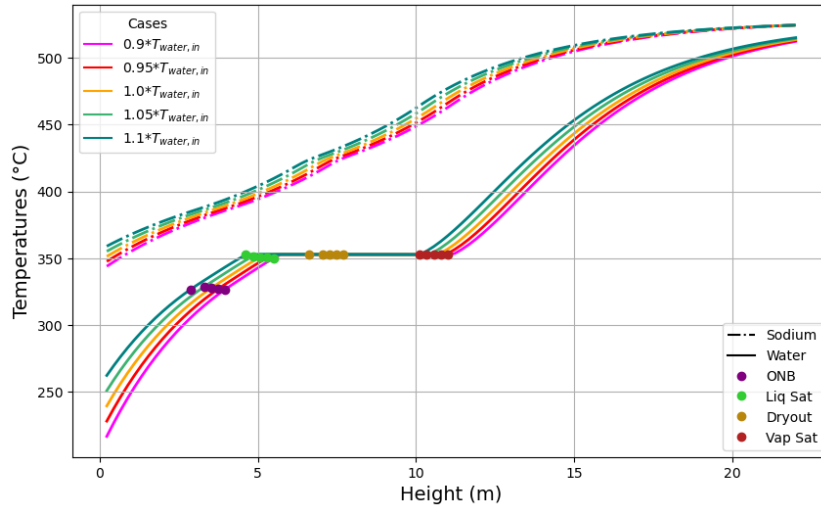


Figure 13: Temperature profiles of liquid sodium and water in case of different water inlet temperatures

Alternatively, heat transfer coefficient does not show any variation.

7.4. Sodium mass flow rate

A larger sodium mass flux is responsible of a higher thermal capacity, which means that more energy is required to make sodium cool down. This leads to higher temperature profiles of both fluids, as shown in Fig.14. Water sees a hotter flow on the other side, so it heats up sooner and reaches its regime transitions earlier along the pipe.

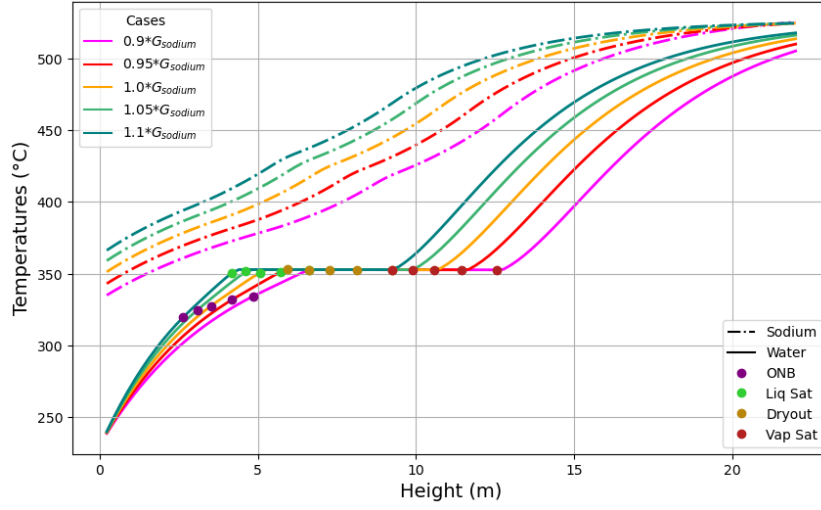


Figure 14: Temperature profiles of liquid sodium and water in case of different sodium mass flow rates

Heat transfer coefficients instead, visible in Fig.15 are not extremely influenced in this particular case. The only notable effect is their increase during the two phase regime due to the enhanced boiling caused by larger temperatures of sodium and consequently of the wall surface as well.

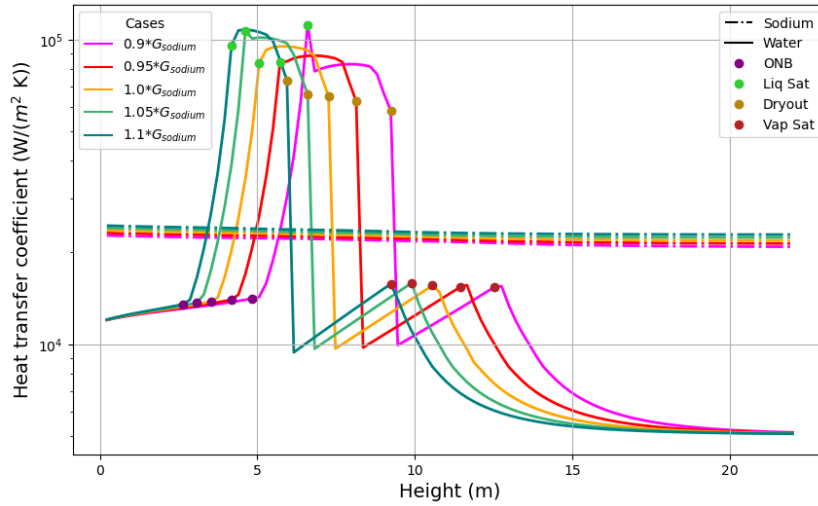


Figure 15: Heat transfer coefficients of liquid sodium and water in case of different sodium mass flow rates

7.5. Sodium inlet temperature

Similarly to the previous case, higher sodium temperatures lead to wider gradients, which causes water to heat up more and sooner. This behavior, presented in Fig.16 is also due to the larger thermal capacity of sodium with respect to water. This implies that the same energy transferred from the former to the latter will cool down sodium not as much as water will heat up.

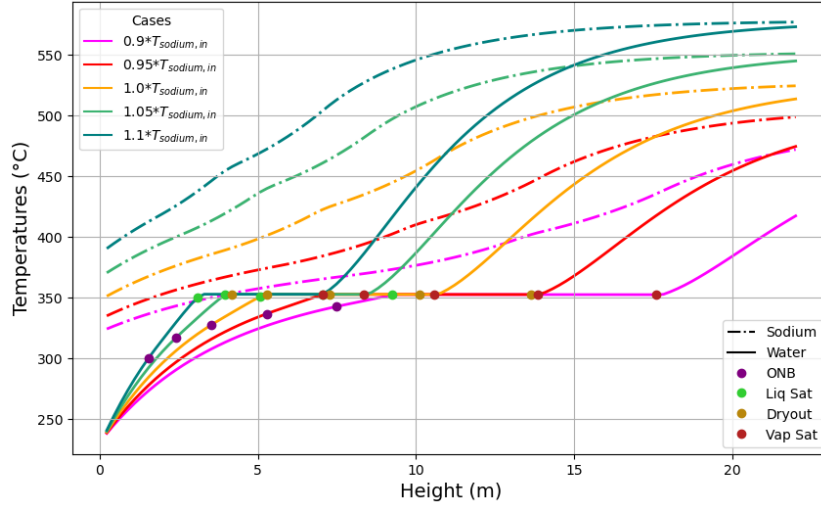


Figure 16: Temperature profiles of liquid sodium and water in case of different sodium inlet temperature

Heat transfer coefficients, visible in Fig.17, are affected in the same way as the prior scenario.

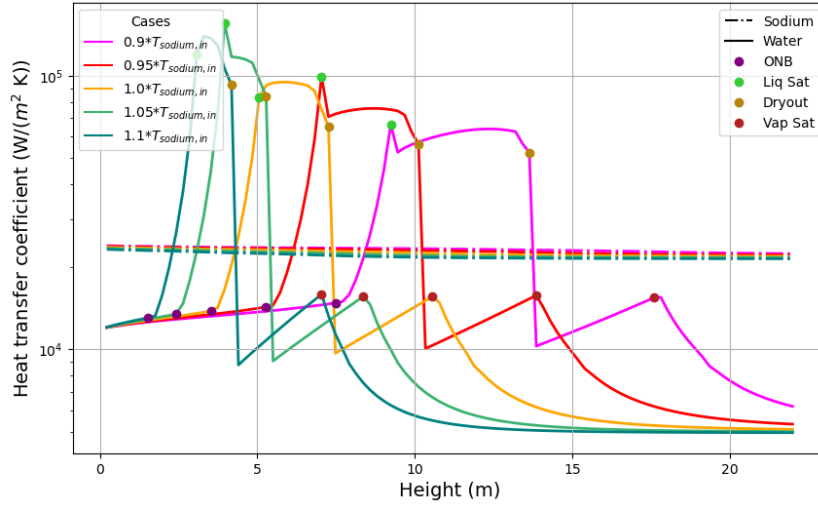


Figure 17: Heat transfer coefficients of liquid sodium and water in case of different sodium inlet temperature

8. Conclusion

To summarize, for the most part, the data calculated through the developed numerical model shows a good agreement with the results obtained by higher fidelity codes [1]. Therefore, the accordance between these two validates the reliability and accuracy of the designed model. A further improvement would be to remove some of the simplifications made in this first analysis and increase the number of control volumes, although this would negatively affect the already demanding computational cost, and so the required time, for the simulation.

Thanks for reading.

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